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E-261 CATHODIC PROTECTION FOR FIELD PIPING

1. SCOPE OF WORK

The work shall include the design, supply, manufacturing, factory testing, delivery to the site, installation, testing and commissioning of an impressed current and sacrificial anodes cathodic protection system in accordance with Part 4, Subsection 1.2 Summary of Work.

The impressed current cathodic protection shall be provided for all buried piping, tank bottoms and other metals buried under ground, which are directly touching the soil.

Certain reinforced concrete structures at the Site may require cathodic protection in addition to the buried piping system where the concrete is exposed to chloride or sulfide ions or similar harmful substances. The need for cathodic protection shall be determined by structure-to-soil potential measurements in accordance with ASTM C 876. These structures will have the reinforcing steel made continuous and connected to cathodic protection test points.

The sacrificial anode system shall be provided for any steel piles or other metal structures not covered in Technical Specification E-260 which are in direct contact with seawater or sea mud.

2. CODES AND STANDARDS

The Contractor shall comply with the requirements of Part 4, Subsection 2.5, Codes and Standards and the following specific standards:

2.1 National

PUIL	:	Peraturan Umum Instalasi Listrik - Indonesia (General Regulation of Electrical Installation Indonesia).
SPLN	:	Standard Perusahaan Umum Listrik Negara.

2.2 British Standard

BS 7361 : Cathodic protection, code of practice for land and marine applications

2.3 American Standards of Testing Materials

ASTM C 876 : Standard Tests Methods for Half-Cell Potentials of Uncoated Reinforcing Steel in Concrete.

2.4 International Electrotechnical Commission

IEC 60076 : Power transformers

IEC 60146 : Semiconductor convertors

IEC 60529 : Degrees of protection provided by enclosures (IP Code)

2.5 National Association of Corrosion Engineers

NACE RP0169 : Standard Recommended Practice for Contract of External Corrosion on Underground or Submerged Metallic Piping System

2.6 National Electrical Manufacturers Association

NEMA MR 20 : Semiconductors Rectifiers Cathodic Protection Units

3. OPERATING AND DESIGN CONDITIONS

The cathodic protection for field piping shall be designed for outdoor operation as a part of combined cycle power plant located in a climate along the seacoast. Refer to Part 4, Subsections 1.1, 2.14, and 2.16 for meteorological, seismic, and environmental conditions at the site.

4. DETAILED REQUIREMENTS

4.1 General

The cathodic protection system for field piping shall have adequate capacity to maintain all piping and other metallic objects in contact with the soil at a potential of not less negative than -0.85 volt with respect to a saturated copper-copper sulphate half cell.

Rectifiers and anodes shall be selected so that they will not be operated above 80% of their rating to allow margin for adjustment, coating deterioration, etc.

4.2 Transformer / Rectifier Units

All transformer/rectifier units shall be equipped with automatic potential control to maintain a constant protected object to soil potential. All rectifiers shall meet the requirements of NEMA MR-20.

The control equipment shall be arranged that each group of equipment shall be controlled independently of other equipment.

Transformer/rectifier units shall be of the self cooled dry type.

Each transformer shall be three-phase, two-winding and self cooled. Dry type transformers shall have class B (or better) insulation to operate within the temperature rise limits for class B insulation. It shall be equipped with step less voltage control, with coarse and fine adjustment, for DC control from 10% to full rated output DC voltage. The voltage shall also be manually operated.

The rectifier shall be a three-phase, full-wave bridge type, suitable for operation from 400 volt, 50 hertz, 3 phase supply (refer to Clause 3.1 Part 4, Technical Specification E-001).

All rectifiers feeding deep-well anode shall be identical. They shall have a provision for controlling the current fed to each anode individually without the use of resistors or external DC control cabinets.

All rectifiers feeding distributed anode beds shall be identical.

The transformer/rectifier shall be equipped with controls to automatically regulate the metal-to-soil potential to within ± 10 mV of its preset voltage to ensure adequate, but not excessive, protective current. The control circuit shall be designed to minimize the error in the reference cell reading caused by the protection current. Connections to the reference cell shall be made by low current leads separate from the main powerleads. Both manual and automatic voltage control shall be provided. The rectifier control circuit shall include a current limiter which shall protect the rectifier against short circuits or open

circuits in the rectifier output circuit.

The transformer/rectifier units shall be equipped with an efficiency filter which shall limit the output ripple to 5% rms of the DC output, or less. In no case shall the ripple exceed a value which is not consistent with long life of platinized anodes.

The transformer/rectifier diodes and silicon controlled rectifiers shall be equipped with fuses and surge suppressors. Surge suppression shall be provided at the rectifier output.

The Transformer/rectifier units shall be equipped with two (2) electrically isolated single pole double throw contacts to indicate low output or rectifier failure. The contact shall be suitable for operating in the 220 volt DC control circuits.

Each transformer/rectifier shall be equipped with a DC voltmeter and ammeter for monitoring the output of the rectifier. The meters of outdoor rectifiers shall be mounted in weather-proof enclosure equipped with a window to allow reading the meters without opening the enclosure.

A convenient means for measuring the structure to soil potential shall be provided.

Each transformer/rectifier units shall have filters or other means to prevent voltage or other distortion being fed back into the AC power supply due to the firing of the silicon controlled rectifier or for any other reason. The filter shall limit the transient and harmonic voltage and currents to not more than ten (10)% of the fundamental.

Each transformer/rectifier units shall be equipped with a three-pole AC circuit breaker at the AC line entrance.

A suitable over-temperature device shall be provided and which will trip the AC input circuit breaker of the rectifier cabinet if the temperature inside the enclosure reaches an unsafe value.

Cable terminals and lugs shall meet the requirement of Part 4, Technical

Specifications E-001 and I-002.

There shall be one (1) DC positive terminal and one (1) DC negative terminal, suitable for single conductor insulated cable with ethylene propylene rubber insulation and a Neoprene or Hypalon jacket. One (1) reference cell terminal for each reference cell used in the system and one (1) structure potential terminal shall also be provided for 1.5 mm² cable. A switch shall be provided to switch between reference cells.

All transformer/rectifier enclosure internal wiring shall meet the requirements of PART 4, Technical Specification E-280.

Each transformer/rectifier enclosure shall be equipped with an outside nameplate in accordance with PART 4, Technical Specification E-280.

Each transformers/rectifier enclosure shall be equipped with the manufacturer's standard nameplate. The nameplate may be mounted internally for indoor rectifiers, but shall be mounted on the exterior of outdoor rectifiers. The name plate shall contain the following information as a minimum:

- (1) Manufacturer's name
- (2) Model, type, and serial number
- (3) Alternating current voltage and frequency
- (4) Number of phases
- (5) Alternating current ampere rating
- (6) Direct current voltage ratings (maximum and minimum)
- (7) Direct current rating

A removable cover or covers shall be provided for access to equipment for replacement and adjustment. The front door shall furnish access to circuit breaker handles, instruments, rectifier DC terminals reference voltage measurement terminals, reference cell toggle switch, manual and automatic control adjustments.

All live parts of 110 volts or over shall be protected from contact with persons opening the front door.

An operating instruction manual, including a wiring diagram of the rectifier

equipment, shall be on the inside of the front door or at another convenient location.

Each transformer/rectifier unit cabinet shall be equipped with a NEMA two hole ground pad (for drilling pattern).

Transformer/rectifier units shall be free standing and self supporting. The degree of protection shall be IPW 55 or better according to IEC 60529.

The rectifier enclosures shall be equipped with windows so that it is possible to read monitoring voltmeters and ammeters without opening the doors. The enclosures shall be paint finished suitable for marine service. Each transformer/rectifier unit shall be equipped with all necessary reference electrodes required for operation.

An appropriate cooling system shall be provided for transformer/rectifier unit.

4.3 DC Control Cabinets

The Contractor shall supply any DC control cabinets required by his design. The control cabinets shall consist of resistors connected to a common source used to control the distribution of current between anode beds or between anodes within a bed.

It should be noted that designs which avoid the use of current balancing resistors are preferred.

Indoor DC control cabinets shall have a degree of protection of IP42 or higher. Outdoor DC control cabinets shall have a degree of protection of IPW55. Each DC control cabinet shall be equipped with a front door to allow adjustment of the resistors.

All resistors shall be of the variable wound type. Each resistor shall have at least six (6) taps for adjustment of the resistance from zero to full resistance. The derating of the resistors due to the enclosures shall be taken into account in determining the size of the resistors. The resistors shall not be operated at over half their rating.

An ammeter shall be connected in series with the output of each resistor. If the

ammeters are located in the same enclosure with the resistors, they shall be mounted in the bottom of the cabinet to prevent their being affected by the heat of the resistors.

4.4 Deep Well Anodes

Renewable deep well anodes shall be used as the primary source of protection current, if applicable. The anodes shall be sized to operate for a minimum of seven (7) years before rejuvenation of the anodes or a major overhaul is required.

The design of the anode system shall be such that it is possible to renew or replace the anodes in the well without the need to drill an additional hole.

All materials used in the deep well anodes, such as those used for the casings, anode support ropes, vent pipes, electrical cables, etc. shall be chosen to resist chlorine gas and other contaminants or reaction products which are present or which would be expected to be formed in the deep well.

The depth of the deep wells shall be as necessary to achieve the required value of resistance while maintaining adequately uniform current distribution.

Each well shall be equipped with a vent pipe to prevent the accumulation of gases which may be formed in the well during operation.

The installation and backfill material in the well (both conductive and nonconductive) shall be such that they facilitate the removal and replacement of the anodes.

Each well shall be filled from the bottom of the hole to a point above the top anode with calcined, fluid petroleum coke. The coke shall have a maximum resistivity of 0.10 ohm-cm and shall be fine enough so that 100% of the partier can pass through a 1.5 mm sieve.

All anode cables shall be terminated at a junction box with a minimum degree of protection of IPW 55 and with a sufficient number of terminals to provide a separate terminal for each anode and each rectifier connection in a well.

4.5 Distributed Anodes

The Contractor shall install distributed anode beds as necessary to provide protection current. The anode beds shall be sized for a thirty (30) years life.

The distributed anodes shall not be installed in an area until all excavation in that area is completed and clearance to proceed with the installation is obtained from the Owner's representative. The Contractor shall repair all anodes and anode wiring which are damaged by subsequent excavation at his own expense.

The anodes shall be high silicon or high silicon chromium cast iron. The anodes shall be factory packaged in a sheet metal container of calcined metallurgical grade coal or petroleum coke breeze of a type suitable for cathodic protection service. The coke breeze shall be compacted in the containers by a vibration compaction process.

The connection end of the anode shall be protected against wear which is premature compared to the rest of the anode.

4.6 Sacrificial Anodes

The sacrificial anodes shall be aluminum alloy or zinc. Alloys containing mercury are not acceptable.

The cathodic protection shall have a design life of thirty (30) years without anode replacement. The cathodic protection system shall provide a sufficient quantity of anodes to provide a protective current of not less than 0.15 ampere per square meter of exposed metal to be protected. The Contractor shall make allowance for deterioration of the protective coating and the resulting increased requirements for protective current in his design.

The Contractor's method of installing the anode shall be subject to the Owner's review. The anode mounting details shall provide adequate strength to prevent the loss of anodes or anode damage due to wave action, etc. The anode shall be located so that they are not subject to damage from forces associated with the berthing and mooring of ships. Provisions shall be made to allow simple replacement of the anodes at the end of their effective life.

4.7 Test Points

All piping and similar underground metallic structures shall be connected to

cathodic protection test points.

The test points shall be weather-proof with a minimum degree of protection of IPW 55 in accordance with IEC 60529.

The test points shall be suitable for applying protection current to the buried structures as well as providing a convenient means of making electrical connections to the pipe for performing corrosion surveys. Cathodic Protection Test Points shall be suitable for current and voltage measurements using portable measurement equipment and reference cells.

An adequate number of similar test points shall be provided to the steel pile system.

4.8 Installation

In case, the Contractor applies the cable trenches between the transformer to anodes, the followings are the minimum requirement:

4.8.1 Earthwork and Backfill

Cable trenches shall be dug deep enough so that cables will not be disturbed by plowing, surface digging, paving or excavation. The depth of the trench shall be approximately one (1) meter.

The minimum width of the trench shall be 300 mm. There shall be at least 25 mm between each cable.

The upper and lower backfill material shall be structural fill granular (SFG) in accordance with Technical Specification S-300. The SFG is a general fill material with a compaction density that is to be built up to grade elevation with suitable moisture control and compaction throughout its placement in accordance with Technical Specification S-300.

Compaction of the backfill shall be with mechanical rammers. Compaction with heavy tractors or sheep's foot rollers is not permitted.

4.8.2 Installation of Cables

Before any cables are installed, the trench shall be cleaned of all coarse material, wood, etc. The earth in the vicinity of the trench shall be cleared of any wood products or any other materials which may attract termites.

The cathodic protection cables shall be installed in a high density polyethylene flexible or equivalent duct system. The duct system shall comply with NEMA TC5. In case rigid steel ducts are provided, the same shall be embedded in concrete. The risers shall be wound with anti-corrosive tape. The duct and cable shall be installed on a bedding layer of compacted lower backfill if there is any debris or coarse aggregate in the bottom of the trench. If there is none, the duct and cable may be installed directly in the bottom of the trench. Each duct and cable shall be laid in the trench with a slight snake configuration so as to provide some slack for future earth settlement.

When more than one (1) layer of cables is being installed, a 230 mm covering of compacted lower backfill shall be installed between layers.

The complete cable trench shall be covered with a marking tape which shall be bright yellow polyethylene tape stenciled "CAUTION - BURIED CATHODIC PROTECTION LINE BELOW". (AWAS-ALUR KABEL PROTEKSI KATODIK).

4.8.3 Installation of Distributed Anodes

Distributed anodes shall be installed in accordance with applicable standard which is subject to the Owner's approval.

The first, last, and each turning anode in a bed shall be marked with concrete block markers. The concrete blocks shall be 300 mm square by 760 mm deep. Each concrete block shall have a sign either mounted on a galvanized strut or mounted directly on the top of the concrete with suitable anchors. The sign shall be fiber glass or similar durable material with a white background approximately 180 by 250 mm. Each marked anode shall be marked with the cathodic protection anode number.

4.9 Tools and Test Equipment

The Contractor shall provide one (1) set of all special tools needed to maintain the cathodic protection system, including the rectifiers. These tools shall be new and shall not be used by the Contractor for the erection and commissioning of the Works. The tools provided shall include a potentiometric voltmeter and two (2) copper-copper sulphate half cells for conducting corrosion surveys.

4.10 Calculations

All design calculations shall be submitted for the Owner's review and approval.

5. PERFORMANCE TESTING AND GUARANTEES

5.1 Shop Tests

Shop tests of the electrical equipment for cathodic protection shall be in accordance with the relevant standard of each equipment.

5.2 Site Test and Corrosion Surveys

After installation, the Contractor shall perform commissioning tests to ensure that the equipment is functioning correctly. The tests shall include measurements of anode currents, rectifier voltage, and protected equipment voltages with respect to a reference half cell.

The Contractor shall perform corrosion surveys for the system to inspect and adjust the cathodic protection system at the time of six (6) months and one (1) year after commissioning to ensure the system continuously to provide effective corrosion protection.

6. QUALITY ASSURANCE/QUALITY CONTROL

Quality assurance/quality control for cathodic protection for field piping shall conform to the requirements of Part 4, Subsection 2.7. The procedures so specified shall include, but not be limited to, the followings:

- (1) Packaging and shipping
- (2) Shop test procedures
- (3) Cleaning and painting

The quality assurance/quality control documentation as specified shall include, but not be limited to, the followings:

- (1) Shop test reports
- (2) All other documentation required by applicable codes and standards.

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